Guilherme B. Pagano

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Education

Federal University of Uberlândia — Nov 2021 — Nov 2026
Bachelor of Science in Electrical Engineering — *Uberlândia, MG, Brazil*

- Applied Mathematics, Algorithms, Machine Learning, Feedback Control Systems, Analog Electronics, Competitive Programming, Statistics, Software Engineering.

Professional Experience

Data Engineer | Software — Sep 2024 — Present
IHM Stefanini — *Belo Horizonte, MG, Brazil (Remote)*

- Responsible for a high-performance client-side solution for Gerdau, designing and developing the PID Modeling and Tuning functionality in the "Loop" product, using Rust and WebAssembly (WASM) to optimize industrial control loops directly in the browser.
- Refactored the prediction model retraining architecture for a project at Vale, implementing event-driven autoscaling in Kubernetes, decoupling the load from other microservices to guarantee execution integrity and result delivery.
- Handled the support and automation of production environments using Kubernetes and ArgoCD, applying GitOps practices to ensure high availability and continuous delivery of applications.
- Developed a Computer Vision Proof of Concept (POC) to identify PPE usage in real-time, demonstrating the solution's viability for increasing industrial safety.

Data Engineering Intern — Jun 2022 — Sep 2024
IHM Stefanini — *Belo Horizonte, MG, Brazil (Remote)*

- Refactored, optimized, and implemented unit tests for APIs and datalake ETLs using Python, Docker, AWS, and Kafka, resulting in increased performance and process reliability.
- Maintained and developed data pipelines for industrial process optimization at Vale, ensuring communication integrity with protocols like OPC UA and technologies such as PIMS, Kafka, APIs, and SQL/NoSQL databases.
- Automated a mass balance calculation process for Ternium, optimizing operational efficiency and result accuracy through Python solutions.
- Developed real-time data synchronization between PIMS and LIMS systems for Boticário, ensuring consistency and information flow using Python, Kafka, and AWS SQS.
- Created and standardized documentation and testing processes for the development team, elevating code quality and maintainability across all projects.

Projects

Ergonomic Mechanical Keyboard (Split Keyboard) — linkedin post

- Designed and built an ergonomic mechanical keyboard from scratch, from conception to functional prototype. The project included PCB design in KiCad, case 3D modeling in FreeCAD for 3D printing, and implementation of the open-source RMK firmware, resulting in a customized and functional hardware solution.

Neural Networks from scratch — github.com/gbPagano/nn_from_scratch

- Developed and validated a neural network library in Python and Rust, implementing the backpropagation algorithm, batch training, layer creation (Linear, Convolutional), and various activation functions from scratch. The architecture's effectiveness was successfully validated through MNIST dataset classification.

Extracurricular Activities

Biorobotics Marathon (MABI) (github.com/gbPagano/mabi) — Dec 2024

- Won 1st place in the competition by leading a project that involved developing a control system for a robotic arm, using an accelerometer and gyroscope-based method for claw movement.

Undergraduate Research in Chess AI — Sep 2024 — Sep 2025

- Implemented algorithms for a chess AI engine, tracking and validating its performance evolution.

- Developed the engine's core components, including a bitboard-based move generator, optimized for maximum computational efficiency and engine performance.

Biofy Hackathon (github.com/gbPagano/hackathon-biofy) Mar 2024

- Led the team that won 3rd place by developing an AI solution capable of classifying and identifying pathogens from sample image analysis.

Teaching Assistant - Procedural Programming Jun 2024 — Nov 2024

- As a scholarship teaching assistant, guided students in developing assignments and projects in C language, reinforcing key concepts such as pointers and memory allocation.

Teaching Assistant - Script Programming May 2022 — Jun 2022

- As a volunteer teaching assistant, offered support to incoming students in learning Python, facilitating their introduction to fundamental programming concepts and computational logic.

Skills

- **Programming Languages**: Python, Rust, Bash, C, SQL.
- **Technologies**: Git, Linux, Docker, Kafka, AWS, Airflow, Kubernetes, ArgoCD, TimescaleDB, FastAPI, Pytest, WebAssembly.
- **Languages**: Portuguese (Native), English.